WSPRadar.org reconstruction & comparison

SNR differences between a short portable vertical and a reference vertical

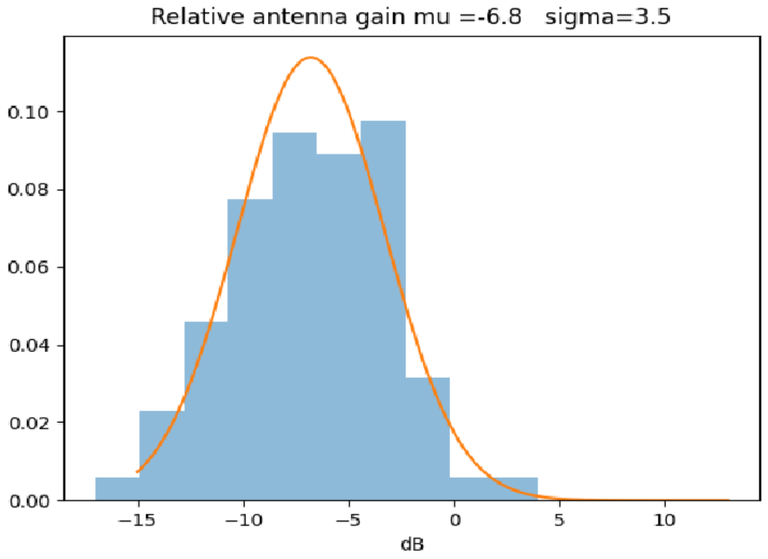
Referenced publication: Zander, J. (2022). *Simple HF antenna efficiency comparisons using the WSPR system.*

Source figure: **Figure 4, p. 4** · arXiv:2209.08989v1 · 19 September 2022

Demo: TX Benchmark | Experiment A | 21 May 2022, 09:30-10:30 UTC | 20 m (14 MHz) | Target SK0WE/P − Reference SK0WE/1 (both JO97) | Reference correction 0.0 dB

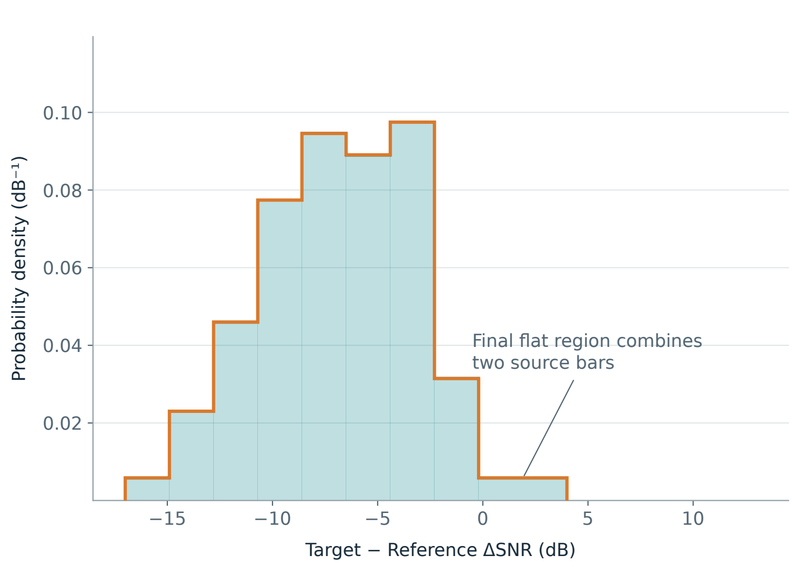
Panel A - Original publication

Original image; orange line: paper's normal curve.



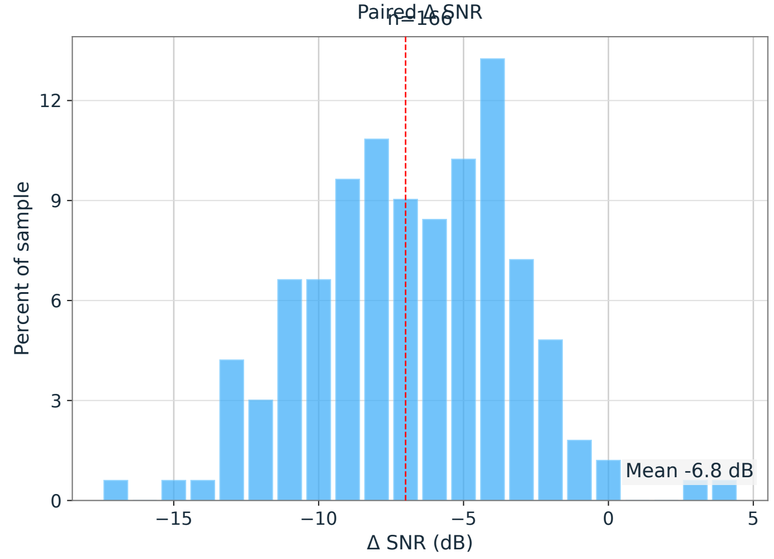
Panel B - Reconstruction

Paper-style bins, with digitized paper bars overlaid.



Panel C - WSPRadar view

Native spot-pair histogram with 1 dB bins.



Rounded paper statistics

Mean −6.8 dB · Reported σ 3.5 dB

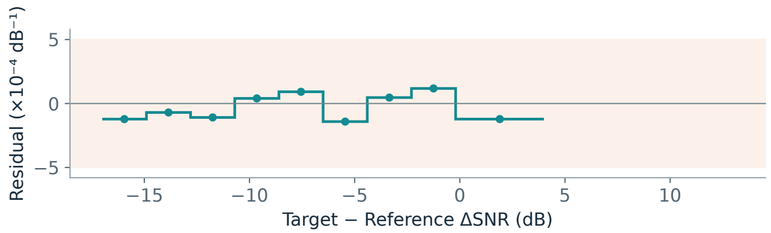
Current WSPRadar reconstruction

Mean -6.8 dB · Sample σ 3.5 dB

166 receiver-cycle pairs · 37 receiver identities · 12 cycles

Agreement check - Panel B minus paper

Maximum |residual| 1.41 × 10⁻⁴ dB⁻¹



Same observations, finer display

The 1 dB bins reveal uneven counts and empty bins that the paper's coarser histogram combines. The mean and standard deviation do not change.

Panel B: density = count / (166 × bin width in dB); bar areas sum to 1. Panel C: percent of all 166 pairs per 1 dB bin; bar heights sum to 100%.

Read both histogram axes linearly. Panel C retains WSPRadar's native binning, colours and statistics.

No fitted curve, time shift or density rescaling is applied to the reconstruction.

READ ΔSNR = Target − Reference: negative values favour the Reference vertical. Each spot pair pairs both signals at one receiver in one cycle.